

Nanda Kishore R

Software Engineer — Backend (Java / Spring Boot)

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Summary

Software engineer with one year of full-time onsite experience on a multi-tenant healthcare platform, working across REST APIs and the PostgreSQL data layer — concurrency control, row-level authorization and transactional integrity. Builds backend services in Java 17 / Spring Boot and Python / FastAPI. Roughly 650 commits, 138 automated test cases and 7 architecture decision records in 12 months.

Skills

- **Languages:** Java (17), Python, SQL, TypeScript, JavaScript
- **Backend & Data:** Spring Boot, Spring Security, JPA/Hibernate, REST APIs, JWT, FastAPI, PostgreSQL, RabbitMQ, Flyway
- **Testing, DevOps & Practices:** JUnit, Mockito, Testcontainers, ArchUnit, pytest, Docker, AWS (EC2), Git, GitHub Actions, CI/CD, Agile/Scrum (Linear), Code Review
- **Concepts & AI:** Object-Oriented Design, Design Patterns, Multi-tenancy, Concurrency, Transactions, Indexing, LangGraph, RAG, Vector Search

Experience

Software Developer Intern (Full-time, 12 months), Botcode (HealthPilot.ai) — Chennai Aug 2025 – Aug 2026

- Built core modules of a multi-tenant clinic platform serving multiple hospitals from one PostgreSQL schema — role-based access control, scheduling, clinical charting and billing — and am the largest contributor to its prescription-pad module (56% of current code).
- Closed a duplicate-record race in clinical chart saves by moving the invariant into PostgreSQL — a partial unique index plus an ON CONFLICT upsert inside one stored procedure — with an idempotent backfill so the constraint could be added to tenants already holding duplicates.
- Hardened an AI clinical-recommendations feature before release: fixed a row-level-security policy that never verified patient ownership, scoped reads to the authoring clinician, enforced organization and role checks on writes, and replaced a read-modify-write settings update with an atomic stored procedure — 40 new tests including a concurrency regression.
- Retired a legacy workflow engine — 116 files and 45 API routes, with 9 tables and 9 functions dropped in one transactional migration — documenting 150 defects and safe-deletion rules in an architecture decision record.

Projects

SkillBridge — Java 17, Spring Boot, Spring Security, JPA/Hibernate, PostgreSQL, RabbitMQ, Testcontainers GitHub

- Designed a multi-tenant training-management API in Spring Boot 3.5 / Java 17 — 20 REST controllers, 122 endpoints, 26 JPA entities — with four-role access control via 113 method-level @PreAuthorize rules and shared-schema tenant isolation enforced by a Hibernate filter plus a 404-not-403 ownership guard.
- Built a transactional-outbox pipeline to RabbitMQ — a SKIP LOCKED lease-based relay, publisher confirms, quorum queues with tiered retries and a dead-letter store with audited replay — verified by killing a real broker: 601 events written before, during and after a five-minute outage were all delivered, none dead.
- Eliminated three N+1 query patterns and locked the fix in with a regression test that fails when a read path's SQL statement count grows with data size (one endpoint went from growing 10 to 28 statements to a constant 5).

Integronix — Python, FastAPI, PostgreSQL, LangGraph, pgvector, Docker, AWS EC2 GitHub

- Built a deterministic, rule-based decision engine (1,025 lines, 41% of the pipeline) scoring candidate medical codes on confidence, documentation-earned specificity, clinical consistency and a negation penalty; fixed an algorithmic-upcoding defect where specificity scaled with raw code length, letting a longer, rarer code outrank a better-documented one — regression-tested both directions.
- Redesigned claim adjudication from a check-then-update race into one PostgreSQL function using an optimistic lock, verified by racing two concurrent approvals against one claim (one success, one 409 Conflict); monetary amounts use Decimal quantized to the cent, and a failed audit-log write triggers a compensating delete rather than leaving an unaudited claim. Backed by a 359-test suite.

Education

Saveetha Engineering College, BE in Electronics and Communication Engineering 2022 – 2026

- CGPA: 8.98 / 10 | Coursework: Data Structures & Algorithms, OOP (Java), DBMS, Operating Systems, System Design

Certifications & Achievements

- **Certifications:** NPTEL Programming in Java (IIT Kharagpur) | AWS Cloud Solutions Architect (Coursera) | Efficient LLM Customization (NVIDIA)
- Published an IEEE conference paper on lightweight deep learning for white blood cell counting (IEEE Xplore); winner, Hardware Track — **AMD Pervasive AI Developer Challenge**; Virtusa hackathon finalist. **GeeksforGeeks Campus Ambassador**; Symposium Coordinator (10+ workshops, 300+ students).